In [13]:
```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split, cross_val_score
from io import StringIO


path = r'C:\Users\Jagatjyoti\Jagat_Code\09-04-2026\archive (1)\movie.csv'
imdb_movie = pd.read_csv(path)
imdb_movie.shape
```

Out[13]:
```
(27278, 3)
```

In [6]:
```
path1 = r'C:\Users\Jagatjyoti\Jagat_Code\09-04-2026\archive (1)\rating.csv'
imdb_rating = pd.read_csv(path1)
imdb_rating.shape
```

Out[6]:
```
(1048575, 4)
```

In [8]:
```
path2 = r'C:\Users\Jagatjyoti\Jagat_Code\09-04-2026\archive (1)\tag.csv'
imdb_tag = pd.read_csv(path2)
imdb_tag.shape
```

Out[8]:
```
(465564, 4)
```

In [11]:
```
print(imdb_tag.columns)
```

```
Index(['userId', 'movieId', 'tag', 'timestamp'], dtype='object')
```

In [12]:
```
print(imdb_rating.columns)
```

```
Index(['userId', 'movieId', 'tag', 'timestamp'], dtype='object')
```

In [14]:
```
print(imdb_movie.columns)
```

```
Index(['movieId', 'title', 'genres'], dtype='object')
```

In [17]:
```
imdb_tag.head()
```

Out[17]:

| | userId | movieId | tag | timestamp |
|---|---|---|---|---|
| **0** | 18 | 4141 | Mark Waters | 2009-04-24 18:19:40 |
| **1** | 65 | 208 | dark hero | 2013-05-10 01:41:18 |
| **2** | 65 | 353 | dark hero | 2013-05-10 01:41:19 |
| **3** | 65 | 521 | noir thriller | 2013-05-10 01:39:43 |
| **4** | 65 | 592 | dark hero | 2013-05-10 01:41:18 |

In [16]:
```
imdb_tag.iloc[0]
```

Out[16]:
```
userId                        18
movieId                     4141
tag                  Mark Waters
timestamp    2009-04-24 18:19:40
Name: 0, dtype: object
```

In [19]:
```
imdb_tag.iloc[1]
```

Out[19]:
```
userId                        65
movieId                      208
tag                    dark hero
timestamp    2013-05-10 01:41:18
Name: 1, dtype: object
```

In [21]:
```
tag_row_0 =imdb_tag.iloc[0]
```

In [23]:
```
tag_row_0
```

Out[23]:
```
userId                        18
movieId                     4141
tag                  Mark Waters
timestamp    2009-04-24 18:19:40
Name: 0, dtype: object
```

In [26]:
```
print(tag_row_0['userId'])
```

```
18
```

In [28]:
```
def greet():
    print("Welcome to ...DS class")
```

In [29]:
```
greet()
```

```
Welcome to ...DS class
```

In [31]:
```
def greet():
    print("Welcome to ...DS class")
greet()
def greet():
    print("Welcome to ...DS class")
      print("Good bye..Boss")
greet()
```

```
Welcome to ...DS class
Welcome to ...DS class
```

In [32]:
```
greet()
greet()
```

```
Welcome to ...DS class
Welcome to ...DS class
```

In [33]:
```
greet()
greet()
greet()
```

```
Welcome to ...DS class
Welcome to ...DS class
Welcome to ...DS class
```

In [35]:
```
def greet():
    print("Welcome to ...DS class")
    print("Good bye..Boss")

greet()
greet()
greet()
```

```
Welcome to ...DS class
Good bye..Boss
Welcome to ...DS class
Good bye..Boss
Welcome to ...DS class
Good bye..Boss
```

In [36]:
```
def add(x,y): # x, y formal argument
    c=x+y
    print(c)
add(4,5) # Actual argument
```

```
9
```

In [37]:
```
def add(x,y):
    c=x+y
    print(c)
add(4)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[37], line 4
      2     c=x+y
      3     print(c)
----> 4 add(4)

TypeError: add() missing 1 required positional argument: 'y'
```

In [39]:
```
def add(x):
    c=x+y
    print(c)
add(4,6)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[39], line 4
      2     c=x+y
      3     print(c)
----> 4 add(4,6)

TypeError: add() takes 1 positional argument but 2 were given
```

In [42]:
```
def add(x,y):
    c=x+y
    print(c)
add(4)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[42], line 4
      2     c=x+y
      3     print(c)
----> 4 add(4)

TypeError: add() missing 1 required positional argument: 'y'
```

In [43]:
```
def add(x,y):
    c=x+y
    print(c)
add(5,6)
```

```
11
```

In [44]:
```
def add(x,y):
    c=x+y
    print(c)
add(5,6)
```

```
11
```

In [45]:
```
def add(x,y):
    c=x+y
    return(c)
add(5,6)
```

Out[45]: 11

In [47]:
```
def add(x,y):
    c=x+y
    return c
add(5,6)
```

Out[47]: 11

In [48]:
```
def add(x,y):
    c=x+y
    return c
result  = add(5,6)
print(result)
print(type(result))
```

```
11
<class 'int'>
```

In [49]:
```
def add(x,y):
    c=x+y
    return c
result  = add(5,6)
print(result)
print(type(result))
```

```
11
<class 'int'>
```

In [50]:
```
def myFun(*args, **kwargs):
    print("Non-Keyword Arguments (*args):")
    for arg in args:
        print(arg)

    print("\nKeyword Arguments (**kwargs):")
    for key, value in kwargs.items():
        print(f"{key} == {value}")

myFun('Hey', 'Welcome', first='Geeks', mid='for', last='Geeks')
```

```
Non-Keyword Arguments (*args):
Hey
Welcome

Keyword Arguments (**kwargs):
first == Geeks
mid == for
last == Geeks
```

In [ ]: